8

ME1100 Thermodynamics

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1st Law of Thermodynamics and PMM1

Natural and man-made changes **must** obey the laws of thermodynamics

Law of energy conservation: first law of thermodynamics

What is PMM1?

Need for a second law of thermodynamics

- **Changes do not take place in any direction**
  - **Water flows down from the hill to the valley. Converse?**
  - **Heat flows from high temperature to low temperature. Converse?**

These are examples of natural spontaneous processes

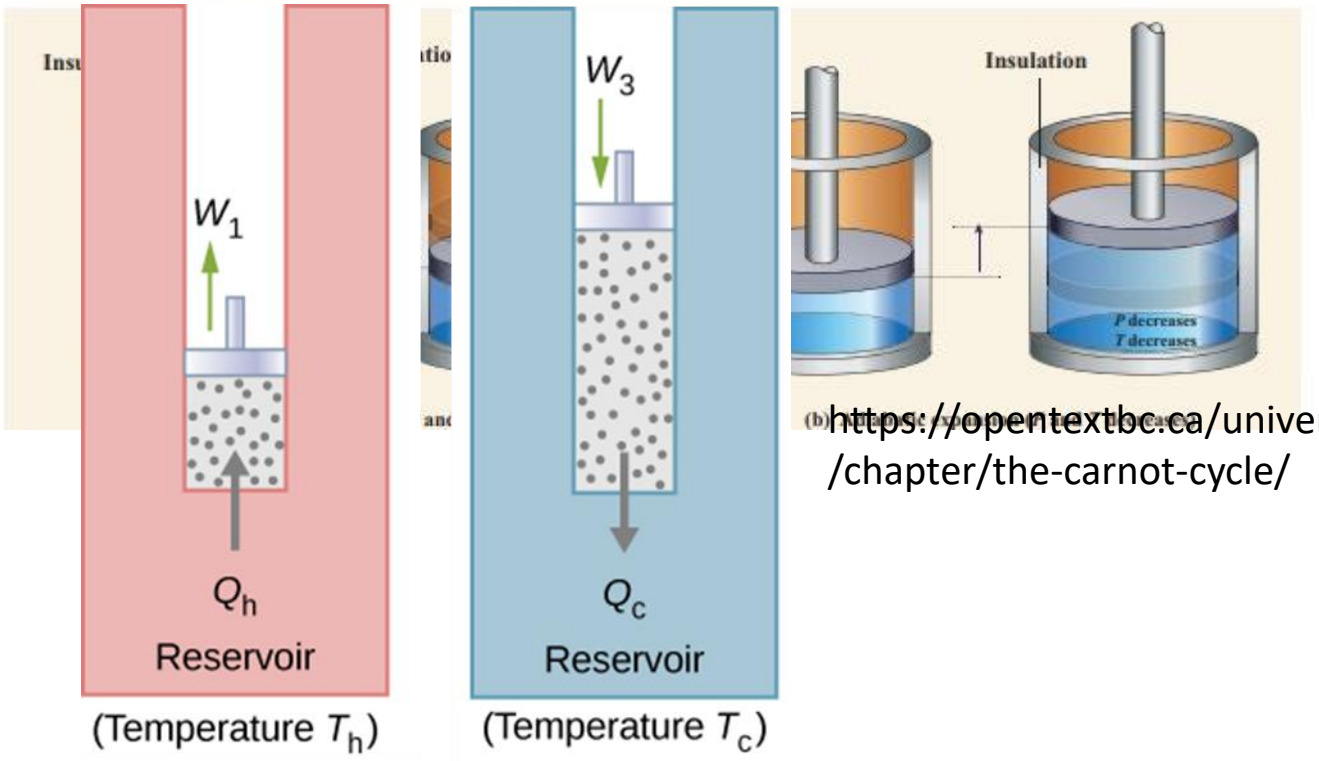
- **Applying brake on a rotating wheel stops motion. Heat is generated.**
- **If the heated wheel is now cooled, will it start rotating? NO**

Examples not restricted to Mechanical Engineering alone

- Chemical reaction will proceed or not
- We grow old and our hair becomes grey

Introduction & Overview

- **Let us consider the expansion (or, compression) of gas. The gas temperature, pressure or volume may be unchanged after expansion or compression.**



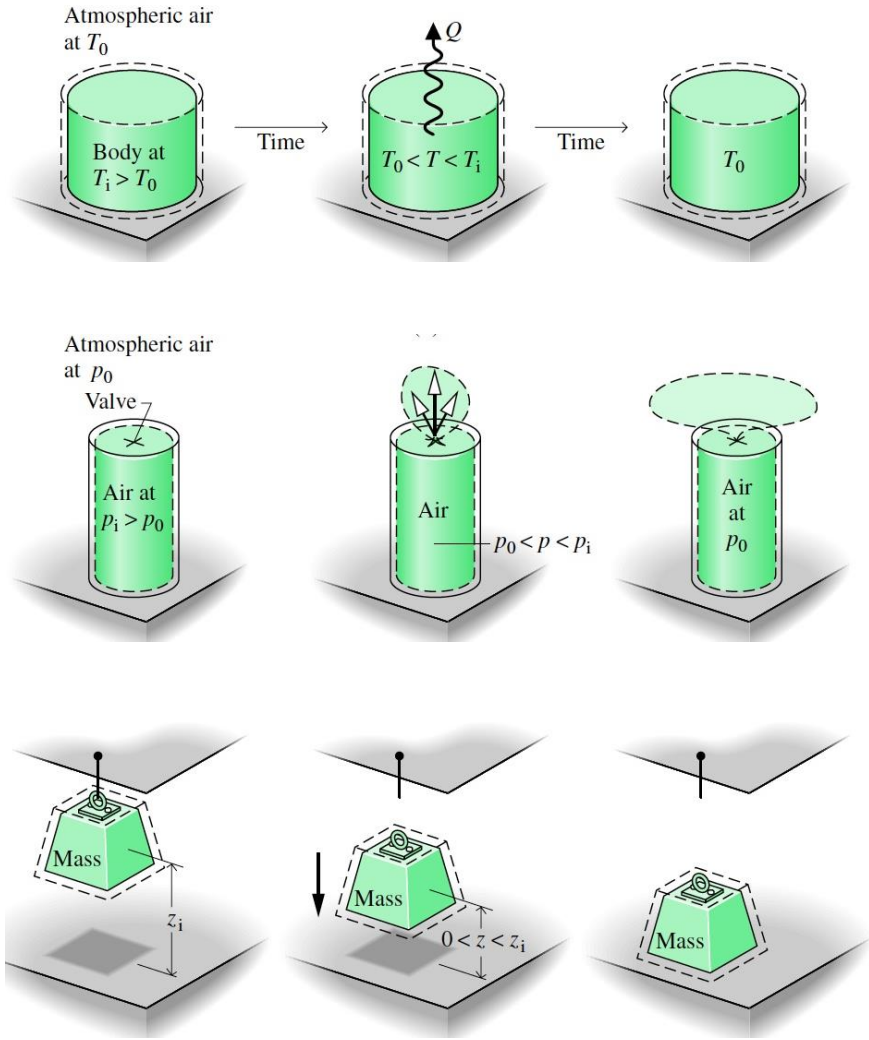
https://www.brainkart.com/article/Adiabatic-process_36250/

https://opentextbc.ca/universityphysicsv2openstax/chapter/the-carnot-cycle/

- **However, there could be changes in the surrounding.**

Introduction to 2nd Law

- First Law talks about conservation of energy in terms of its magnitude and not about the direction of energy flow
- Inverse of processes shown in the Figure (from Moran & Shapiro) cannot occur spontaneously
- These examples also indicate opportunities to extract useful work



Introduction & Overview

- **Conversion of energy is not efficient in all directions**
  - **Generator/Motor: 100% conversion in ideal case?** **YES**
  - **Heat to work: Is 100% efficiency in conversion possible?** **NO**
  - **Work to heat: Is 100% conversion possible?** **YES**
- **Second law: directionality and puts limits on the conversion of energy from one form to another**

Why Second Law?

- Therefore, a framework was needed to examine ideas (to save the trouble of finding that a contraption does not work the way the inventor thought it would). This motivated the need and evolution of second law.

- Heat engines is an abstract concept that is the building block for the second law.

- Before going to the definition of Heat engine, the concept of a thermal reservoir is introduced.

Thermal Reservoirs

- Thermal Reservoir is a large body of matter (in relative proportion to any system under consideration) that has infinite heat capacity.
- That is, any amount of heat transacted with the reservoir does not lead to change in its temperature.
- Examples are ocean, lake, river, ambient atmosphere, 2-phase materials, temperature controlled furnaces, etc.
- Thermal reservoir that supplies heat is called **source**, and that which absorbs heat is called **sink**.

Heat Engines

- Heat engine is a continuously operating device comprising a thermodynamic system operating in a thermodynamic cycle with heat and work interactions. Heat interactions are with thermal reservoirs (source and sink)

- What is not a heat engine?
  - One that does not operate continuously
  - One that does not operate in a thermodynamic cycle
  - One that does not involve both heat AND work interactions

- For continuous operation, cyclic process is a must

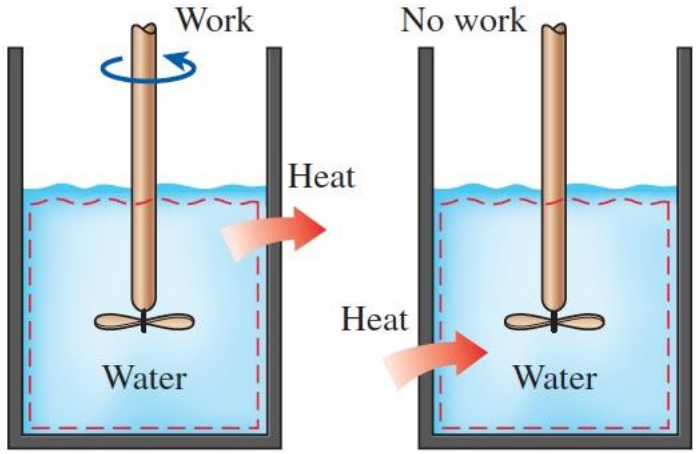
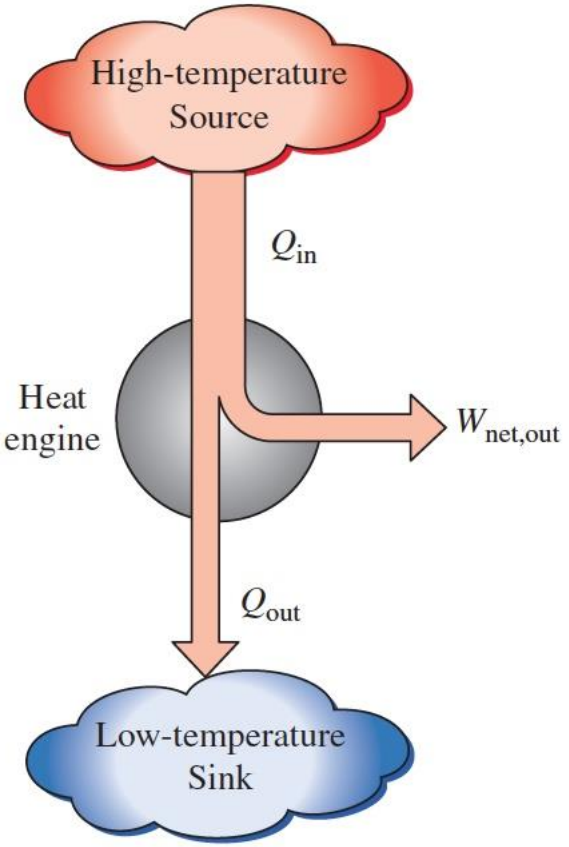


FIGURE 6–8
Work can always be converted to heat directly and completely, but the reverse is not true.

Heat Engine Layout

- Heat Engine is an abstract concept that can be represented by a simple block diagram.

- The **heat engine** receives heat from the source, converts part of it into work, and rejects the remaining to the sink.

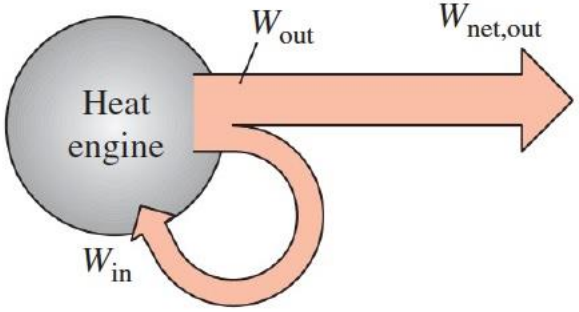


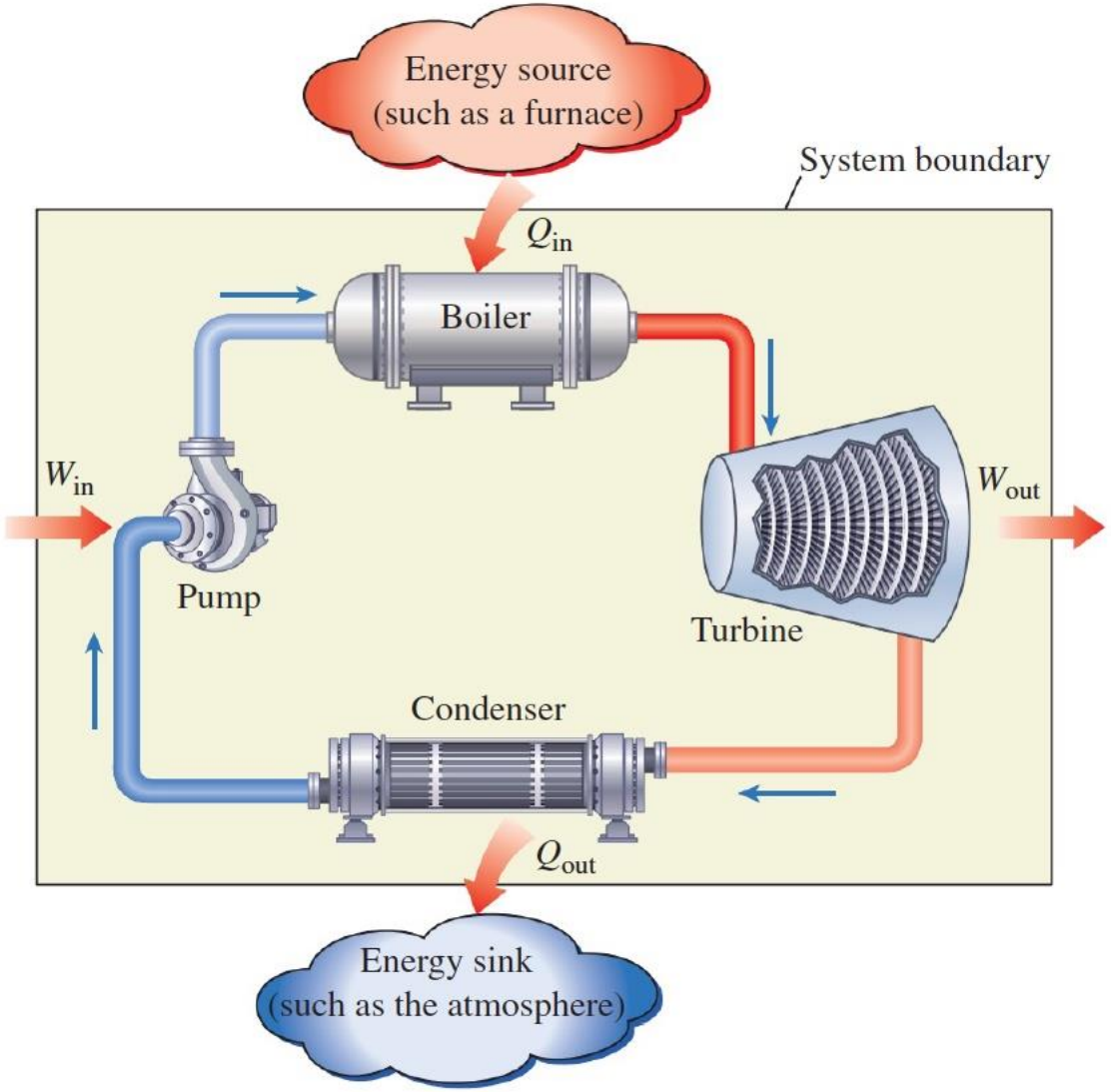
Abstract concept of Heat Engines

- For the analysis of heat engines, no details are required
  - No details of the thermodynamic cycle are required.
  - No details of the engine material, its construction, shape, etc.
  - No details of the thermal reservoirs, **source and sink**.
    Only their **temperatures** are required to be specified.
  - No details about the nature of heat transfer.
    Only the **quantities of heat transfer** are required.
  - No details of the nature/mechanism of work produced is required.
    Only the **work quantity** is required.
- With only the required variables/quantities specified above, one can analyze a heat engine.
- For logical analysis, a simple block diagram representation of the heat engine is sufficient.

Example of practical application of heat engine

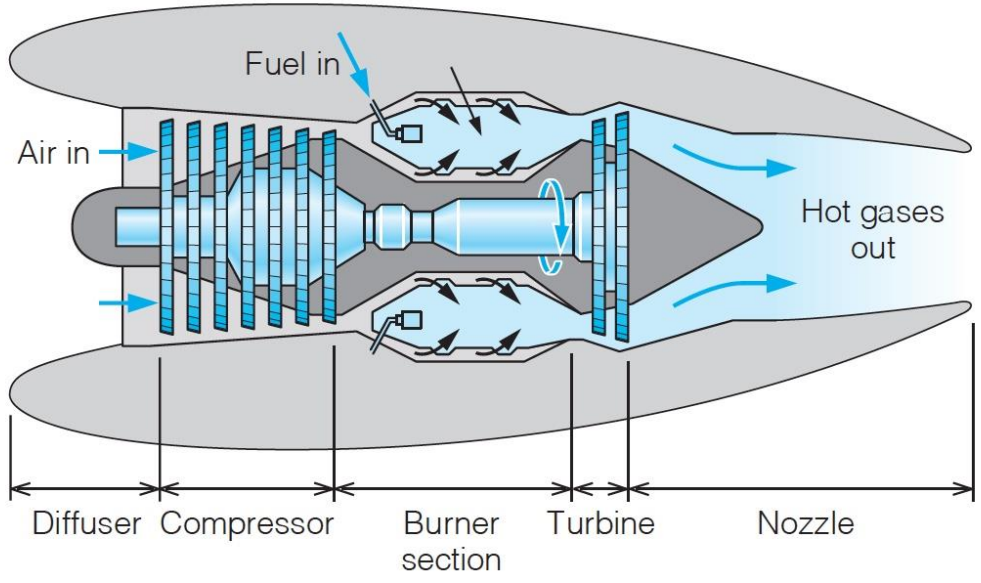
- Thermal Power Plant
- Source is the furnace
- Sink is ambient water body or atmosphere
- Part of the work output from turbine used to run the pump

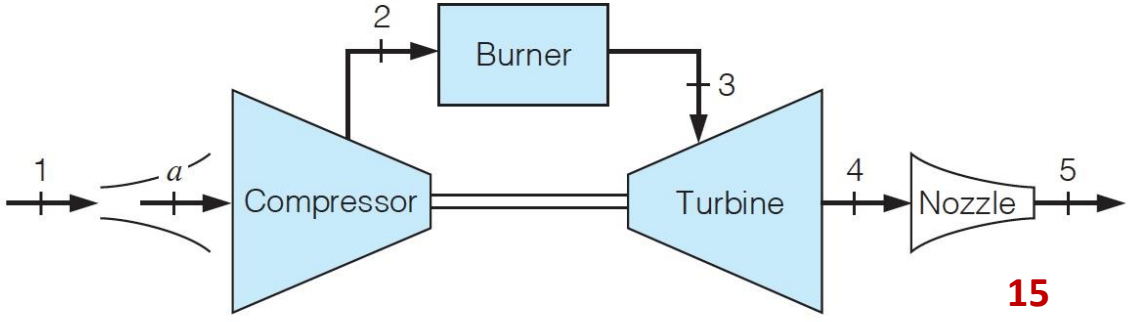




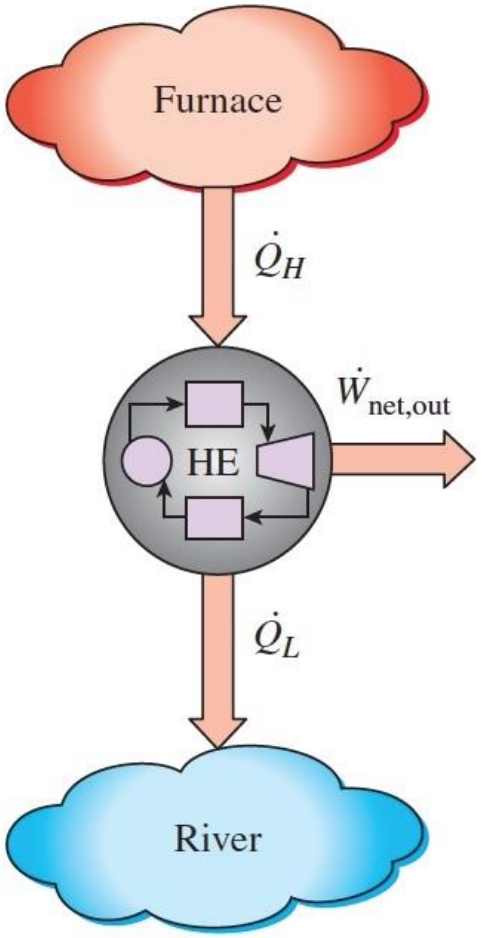
Other examples

- In IC engines , aircraft engines, cycle is not actually completed, but assumed to be completed externally.
- Moreover, the working fluid does not remain the same. Fuel is added to air and becomes combustion products.
- However, approximations can be made to analyze them as heat engines.

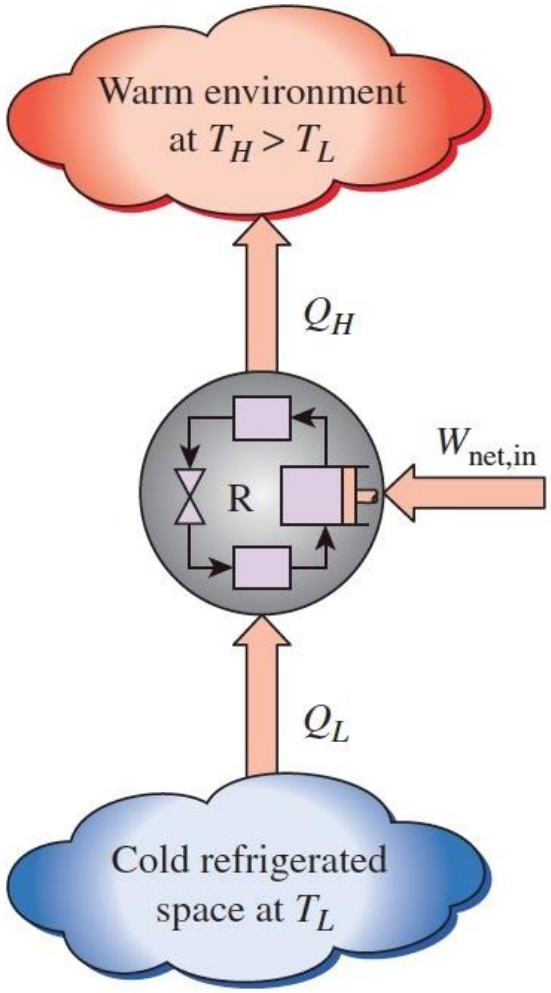




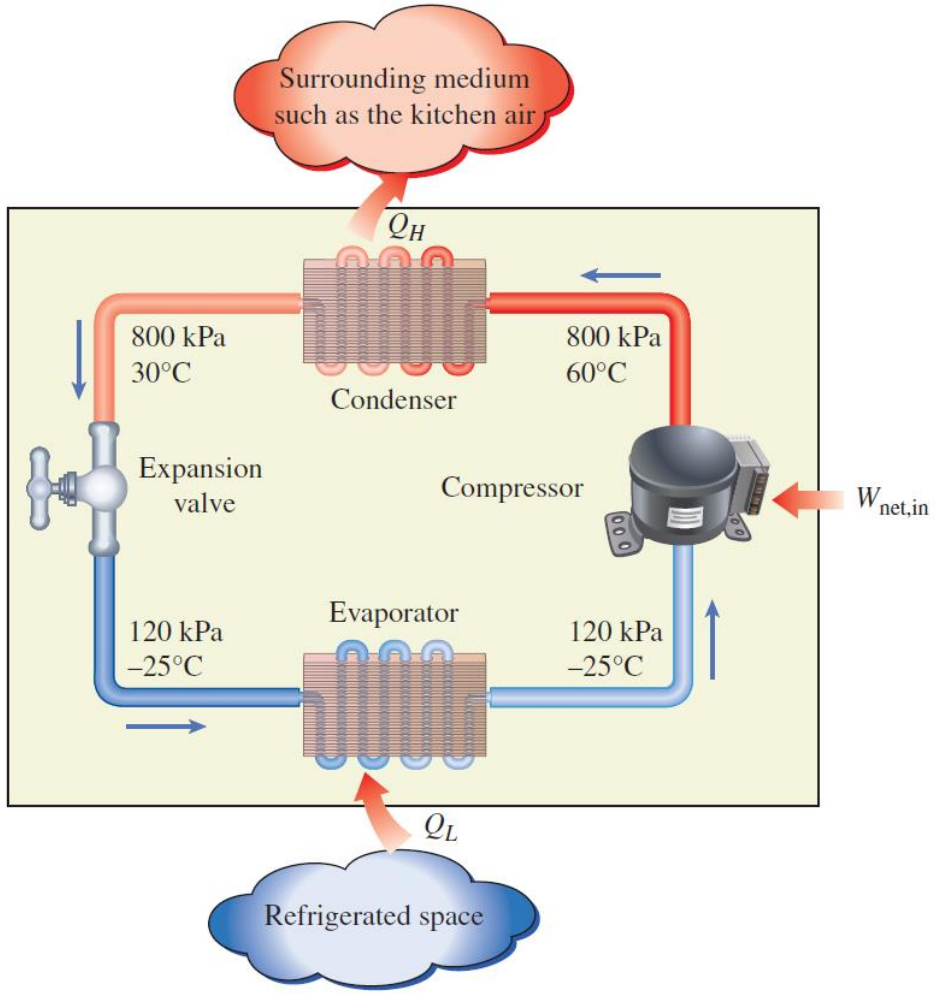
Direct Engines & Reverse Engines



- Conversion of Heat into Work – Direct Engines
  Eg., Power Plants, Auto/Aircraft Engines
- Using work to transfer heat from cold bodies to hot ones – Reverse Engines
  Eg., Refrigerators, Air-conditioners
- Block diagram looks similar, except that the arrows are reversed.



Refrigerators and Heat Pumps



The transfer of heat from a low-temperature medium to a high-temperature one requires special devices called **refrigerators**

The working fluid used in the refrigeration cycle is called a **refrigerant**.

Quantifying the performance of heat engines

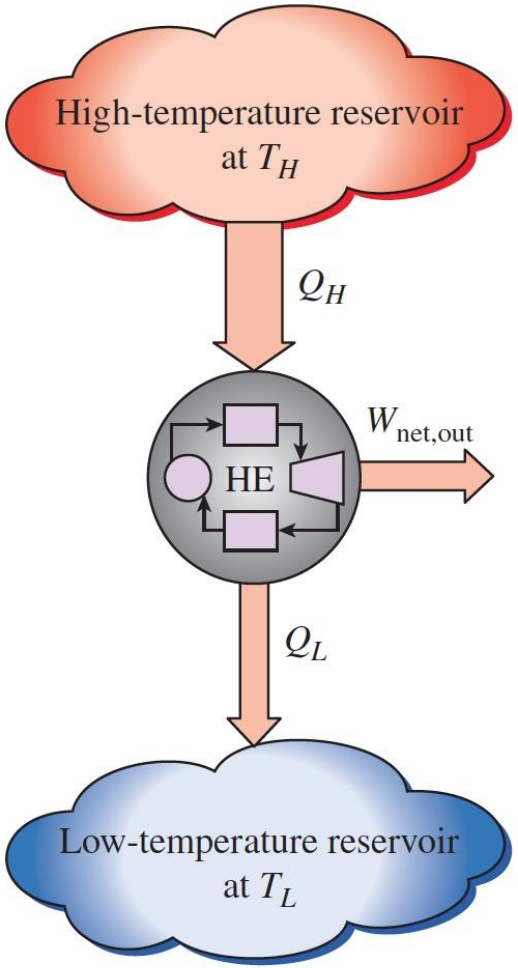
- Thermal efficiency is ratio of the quantity we want to maximize (the work output) to the quantity we want to spend the least (heat input)

$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

$$\eta_{th} = \frac{W_{net,out}}{Q_{in}}$$

$$W_{net,out} = Q_H - Q_L$$

$$\eta_{th} = \frac{W_{net,out}}{Q_H} \quad \text{or} \quad \eta_{th} = 1 - \frac{Q_L}{Q_H}$$



Quantifying the performance of reverse engines (refrigerators, heat pumps)

- One would like to maximize heat transferred from
  - the lower temperature reservoir (refrigerator)
  - the higher temperature reservoir (heat pump)

- One would like to minimize the work spent for achieving this

- Since the ratio can exceed 1, we do not call it efficiency, but term it as “Coefficient of Performance” (COP)

- COP_{Ref} and $COP_{Heat\ Pump}$

COPRef and COPHP

$$COP_{Ref} = \frac{Desired\ Output}{Required\ Input} = \frac{Q_L}{W_{in}}$$

$$COP_{HP} = \frac{Desired\ Output}{Required\ Input} = \frac{Q_H}{W_{in}}$$

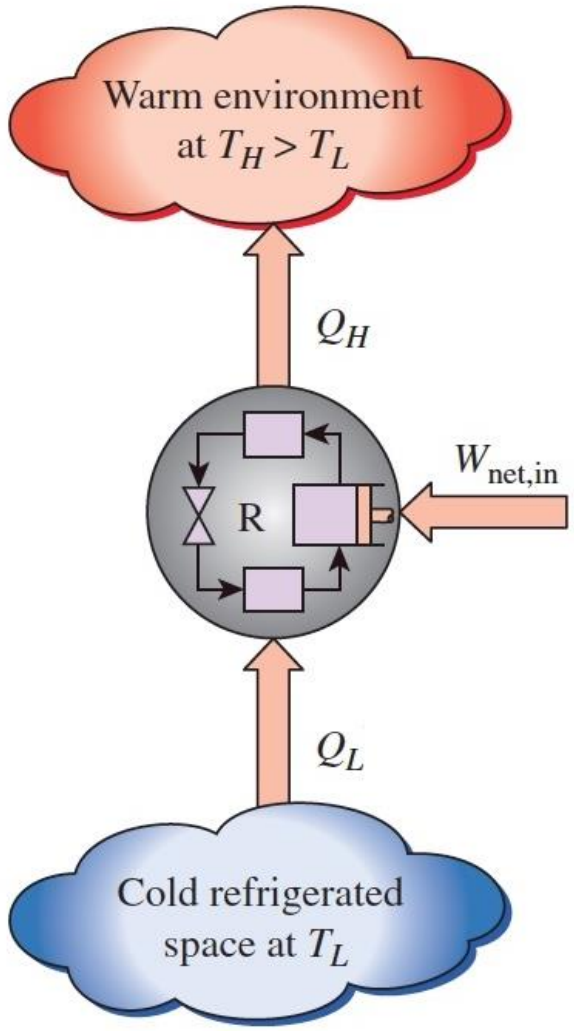
$$W_{in} = Q_H - Q_L$$

$$COP_{Ref} = \frac{Q_L}{Q_H - Q_L}$$

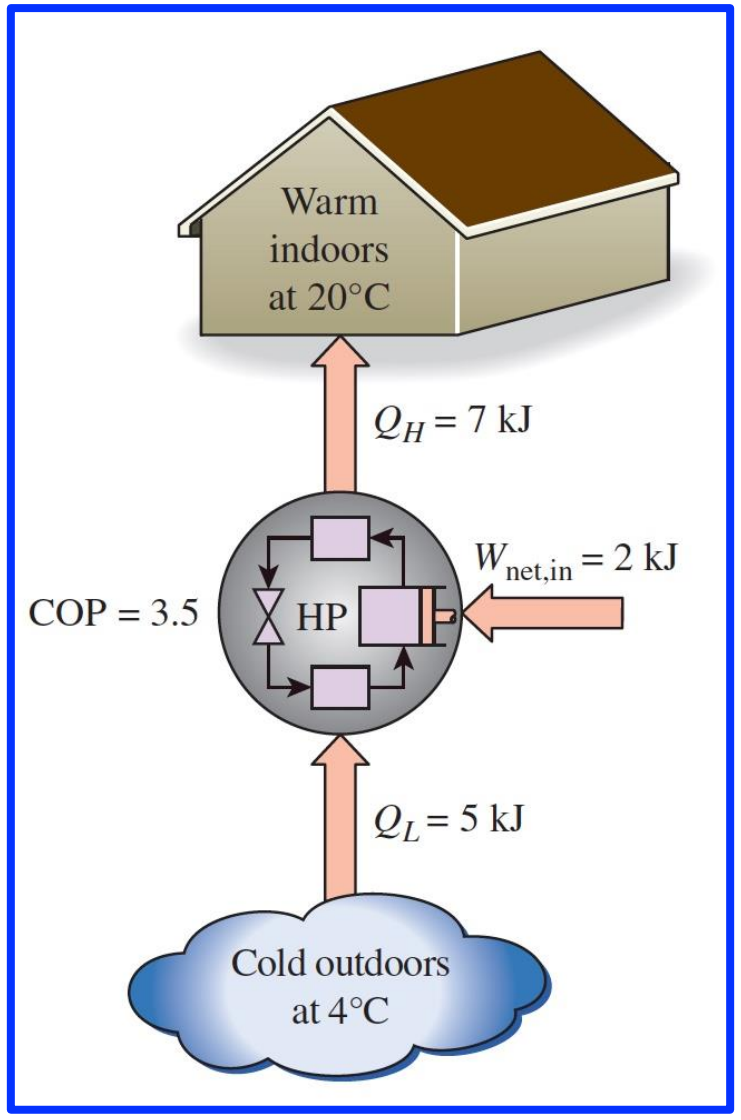
$$COP_{HP} = \frac{Q_H}{Q_H - Q_L}$$

$$COP_{HP} = 1 + COP_{Ref}$$

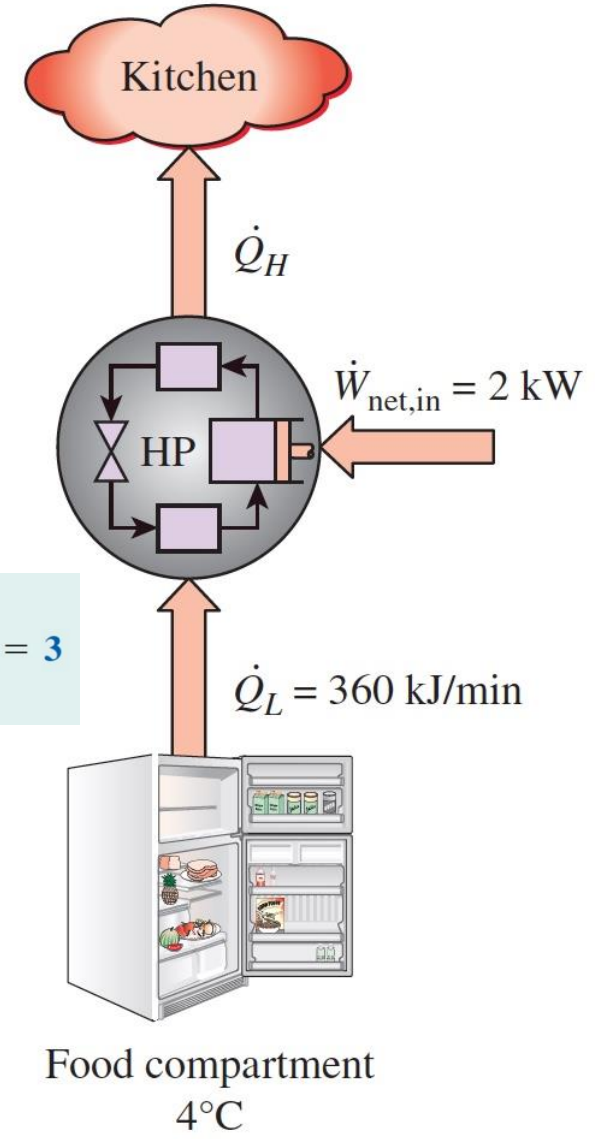
In the block diagram of the reverse engine, when we are interested in Q_H (eg., room heating), we call it **heat pump** and when we focus on Q_L, we call it **refrigerator**



Examples of Heat Pump and Refrigerator



$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{net,in}} = \frac{360 \text{ kJ/min}}{2 \text{ kW}}\left(\frac{1 \text{ kW}}{60 \text{ kJ/min}}\right) = 3$$



Second Law Statements

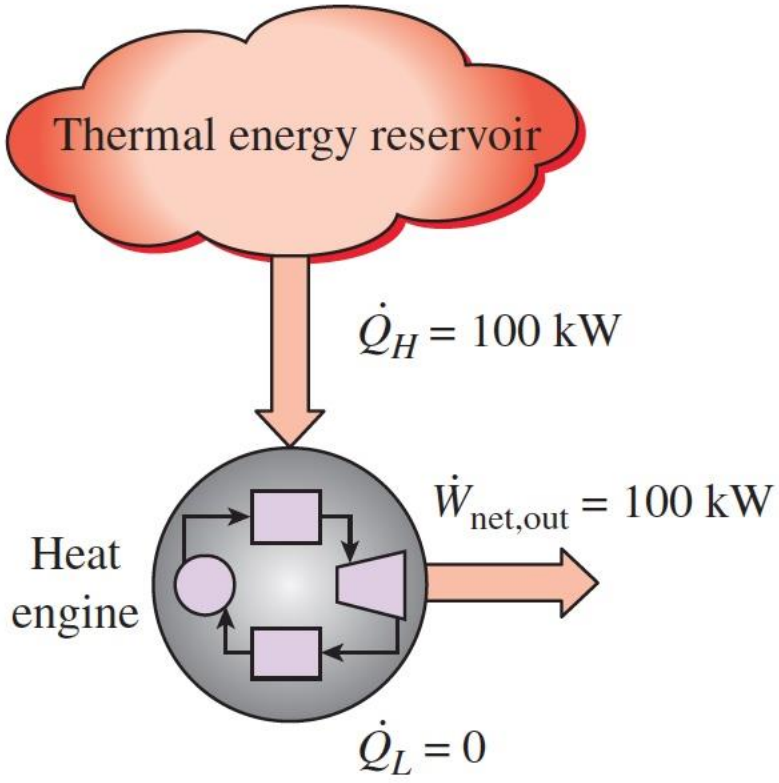
- Having understood the concept of heat engines, we are equipped to study second law
- Independently stated by Lord Kelvin, Max Planck and Rudolf Clausius
- Two equivalent statements (i) Kelvin-Planck Statement and (ii) Clausius Statement
- Similar to First Law, there is no proof, but continued to be in force since all observed events obey them and since there is no evidence otherwise

Kelvin-Planck Statement

- It is impossible to construct a device that will operate in a thermodynamic cycle and produce a net amount of work by exchange of heat with a single reservoir

- That is, two reservoirs are required, the second one being the sink where some heat is to be rejected

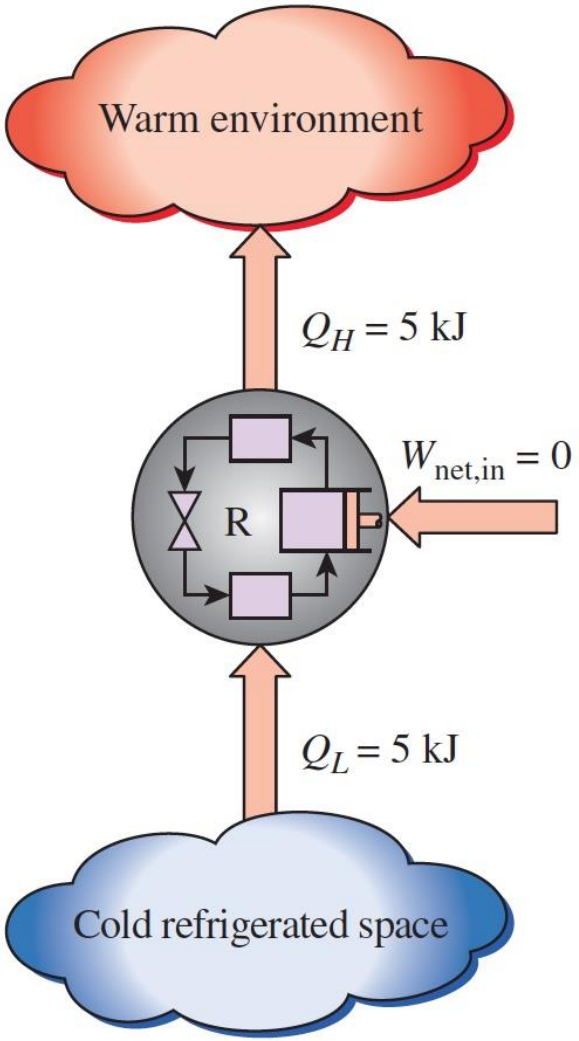
- It implies, thermal efficiency of 100% is unachievable (even ignoring dissipative, effects like friction, viscosity)



Clausius Statement

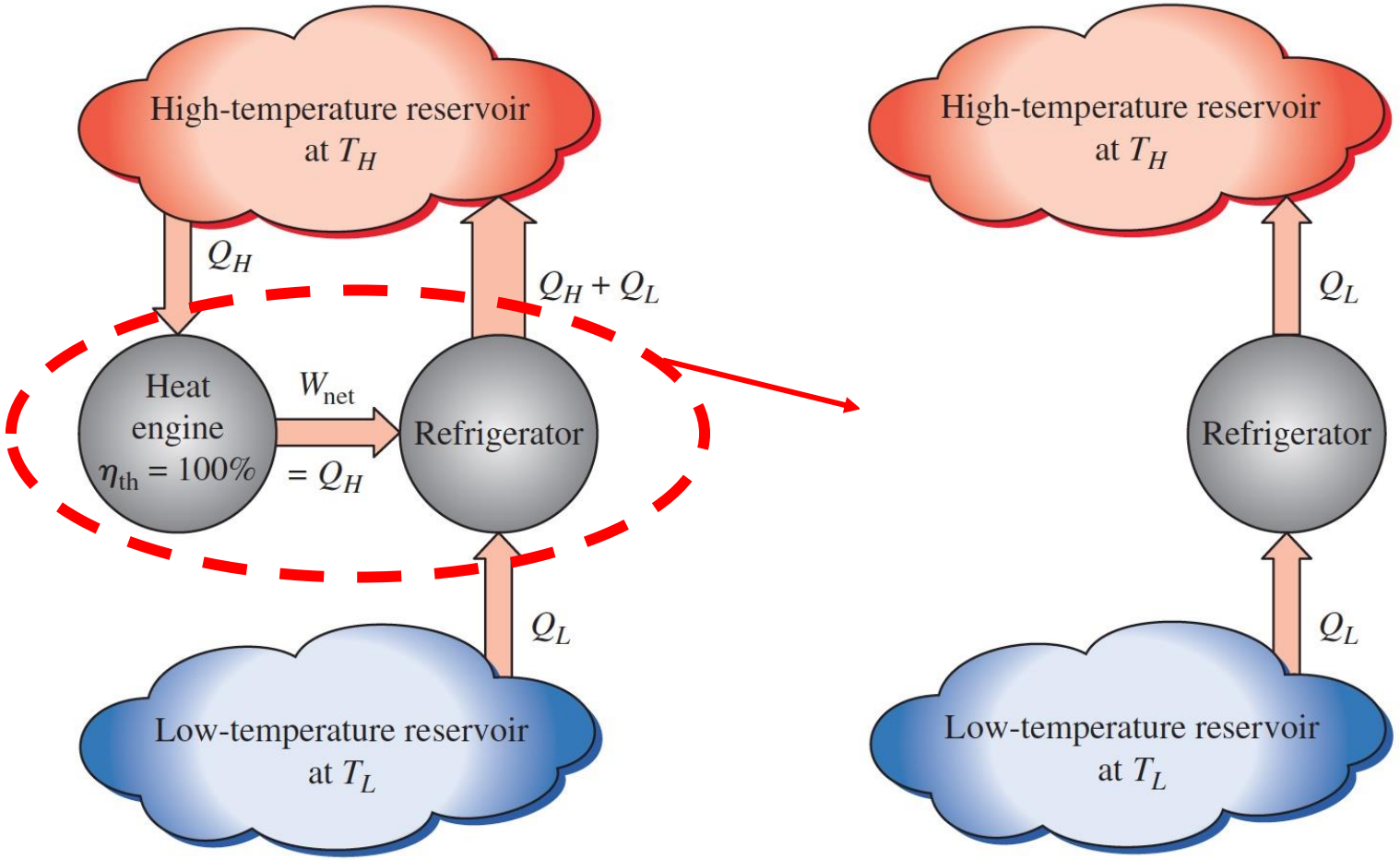
- It is impossible to construct a device that operates in a thermodynamic cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body

- While K-P statement refers to direct engines (heat engines), Clausius statement pertains to reverse heat engines (heat pumps and refrigerators)



Kelvin-Planck and Clausius Statements are equivalent

Violation of Kelvin-Planck statement leads to the violation of Clausius statement also!

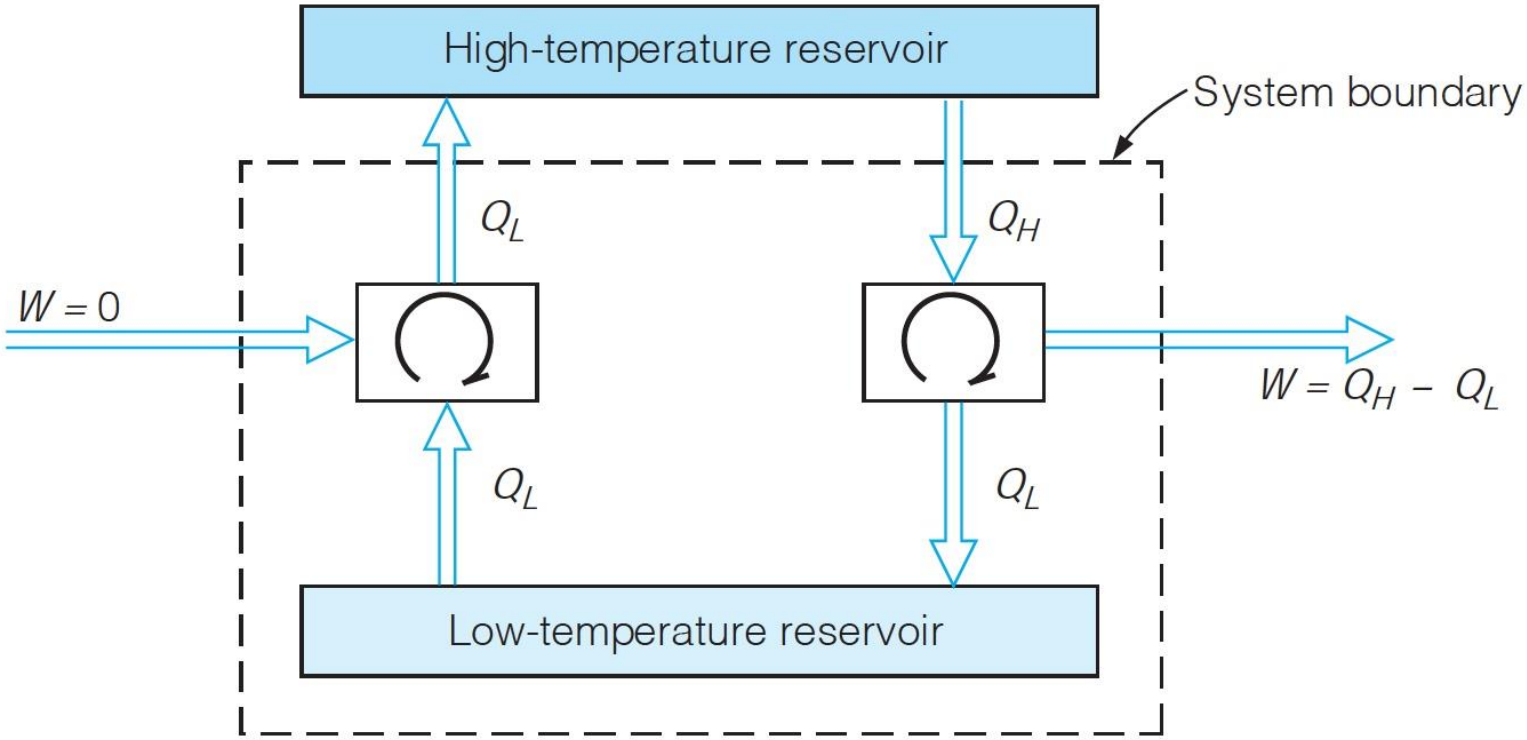


(*a*) A refrigerator that is powered by a 100 percent efficient heat engine

(*b*) The equivalent refrigerator

Another example

- Violation of Clausius statement leads to the violation of Kelvin-Planck statement too!



- Hence, the two statements are equivalent

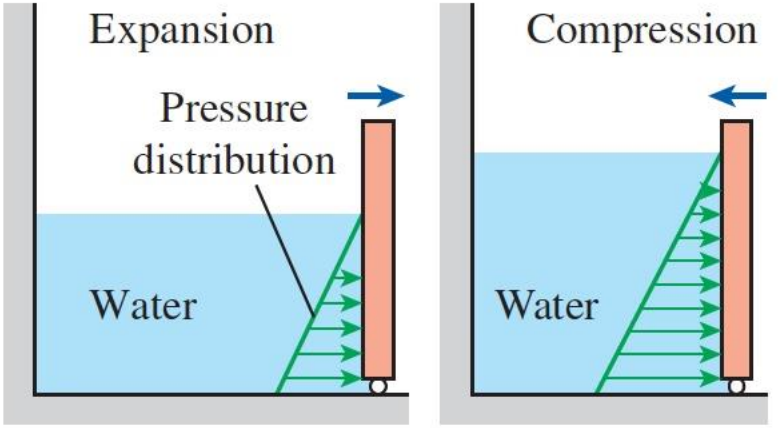
Reversible and Irreversible Processes

- Since we know that we cannot attain 100% efficiency, we come to the next question, “what is the best efficiency one can get?”
- To answer this, we need to learn the concept of reversible and irreversible processes.
- A reversible process is defined as a process that retraces its path if the cause is reversed, without leaving any trace in the surroundings.
- This is possible when the process is a quasi-equilibrium or quasi-static process.
- So, all intermediate states are traversed by the system during a reversible process. Therefore, we denote the process using a solid line. In a irreversible process, the actual path is unknown.

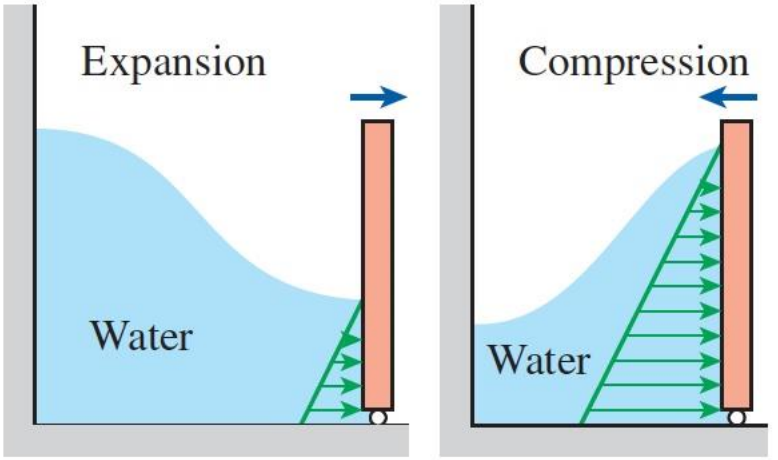
Nature of Reversible processes

- Reversible process delivers the maximum amount of work and consumes the least amount of work

- Irreversible process delivers lesser work and consumes more work

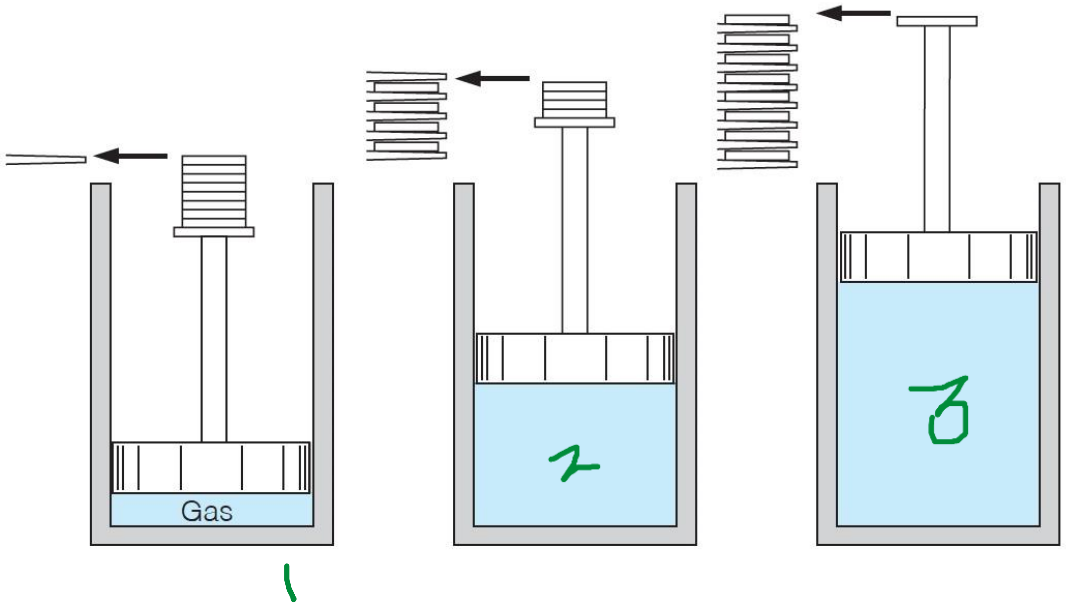


(*a*) Slow (reversible) process



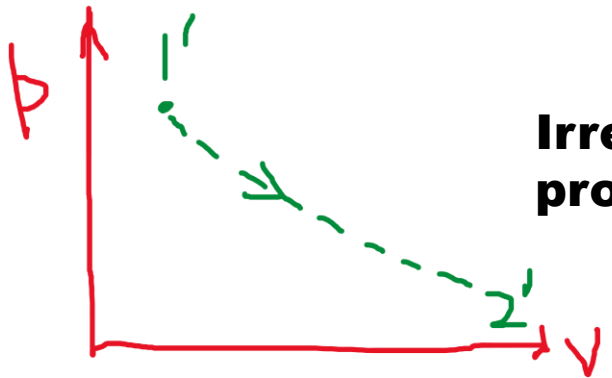
(*b*) Fast (irreversible) process

Another example

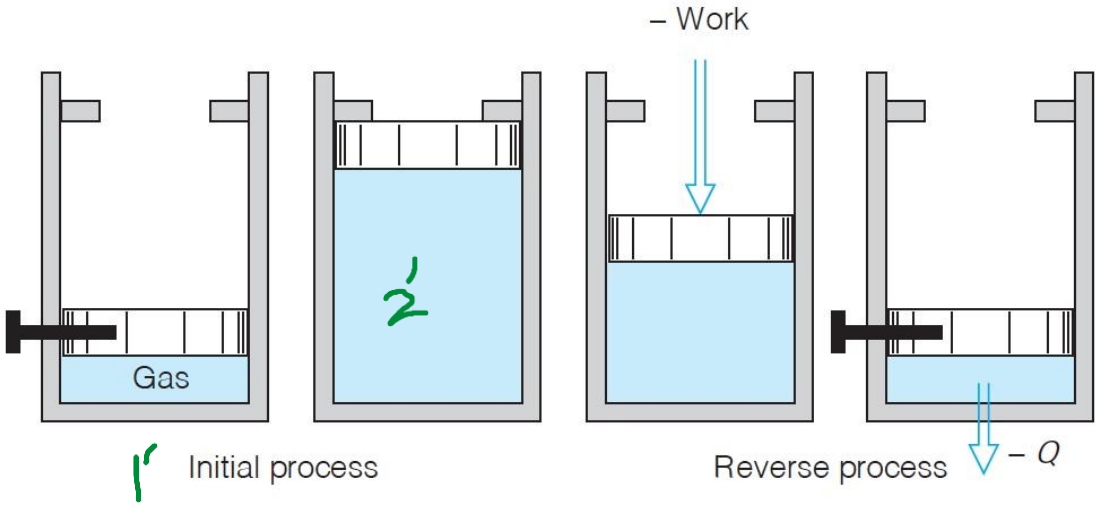


Reversible process





Irreversible process



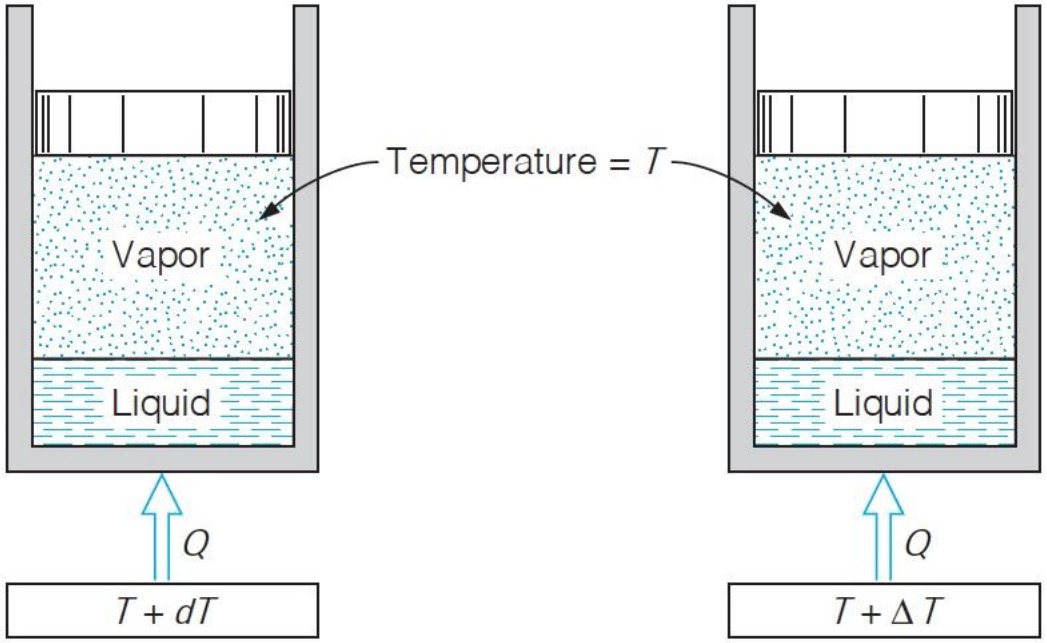
Causes of Irreversibility

- Non-Equilibrium processes
  - ❑ Unrestricted/partially resisted expansion
  - ❑ Heat transfer through finite temperature gradient
  - ❑ Combustion/Chemical reaction
- Dissipation
  - ❑ Friction (solid-solid),
  - ❑ Friction in solid-fluid, fluid-fluid (viscous skin-friction and pressure drag)
  - ❑ Plastic deformation of materials, Hysteresis effects
  - ❑ Resistance to flow of electricity
- Mixing of two or more substances
  - It takes energy to separate substances that are mixed

Internal and external irreversibilities

- Irreversibilities occurring within the system boundary are called internal irreversibilities, and those outside the boundary are external irreversibilities.

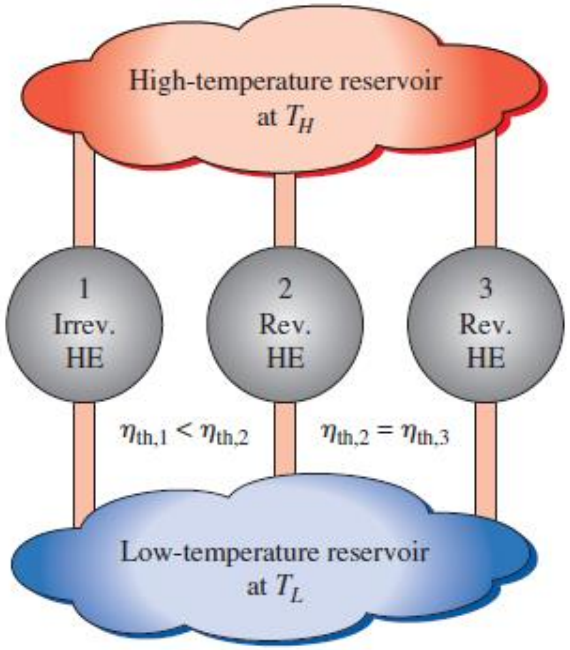
Both internally and externally reversible

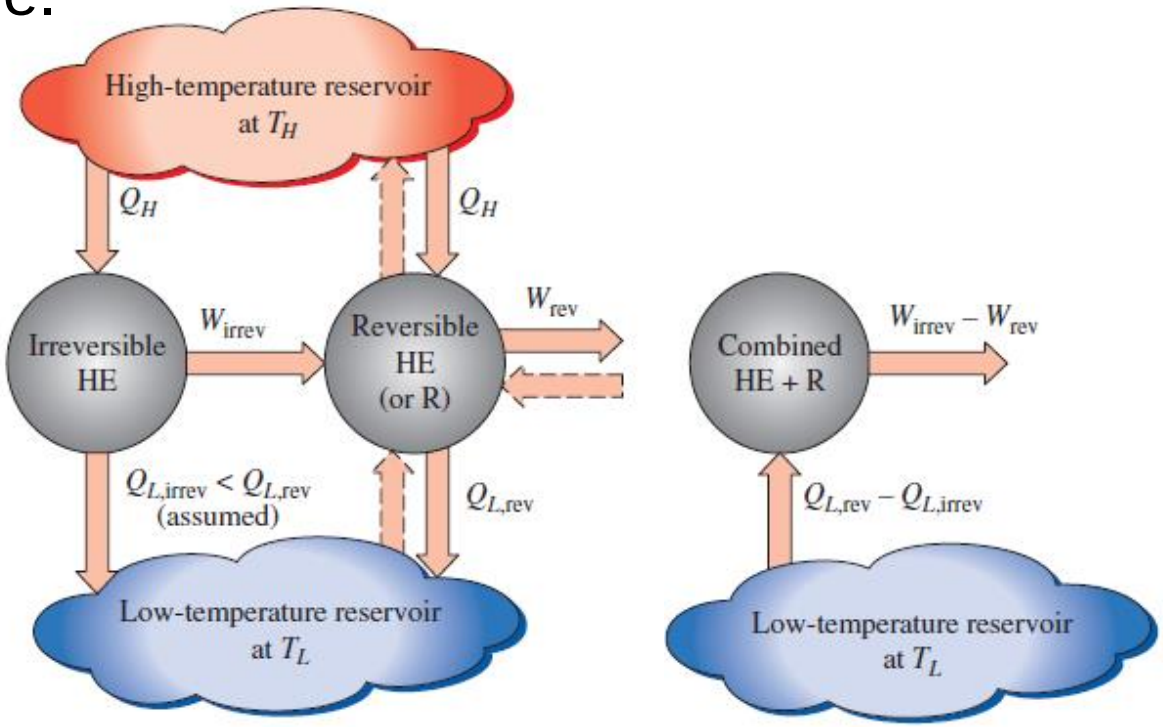


Internally reversible but with external irreversibility (heat transfer with finite temperature gradient)

The Carnot Principles

- The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs
- The efficiencies of all reversible heat engines operating between the same two reservoirs are the same.



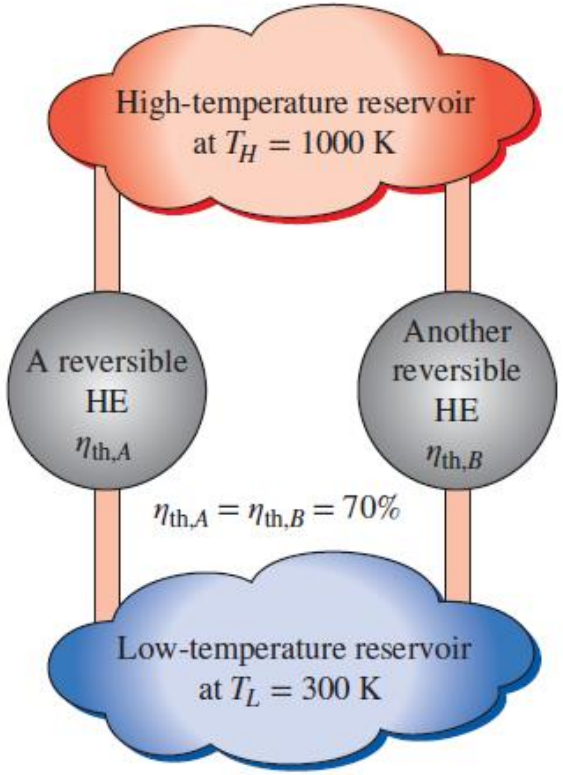


(*a*) A reversible and an irreversible heat engine operating between the same two reservoirs (the reversible heat engine is

(*b*) The equivalent combined system

Absolute temperature scale

All **reversible heat engines** have the **same thermal efficiency** when operating between the **same two reservoirs**

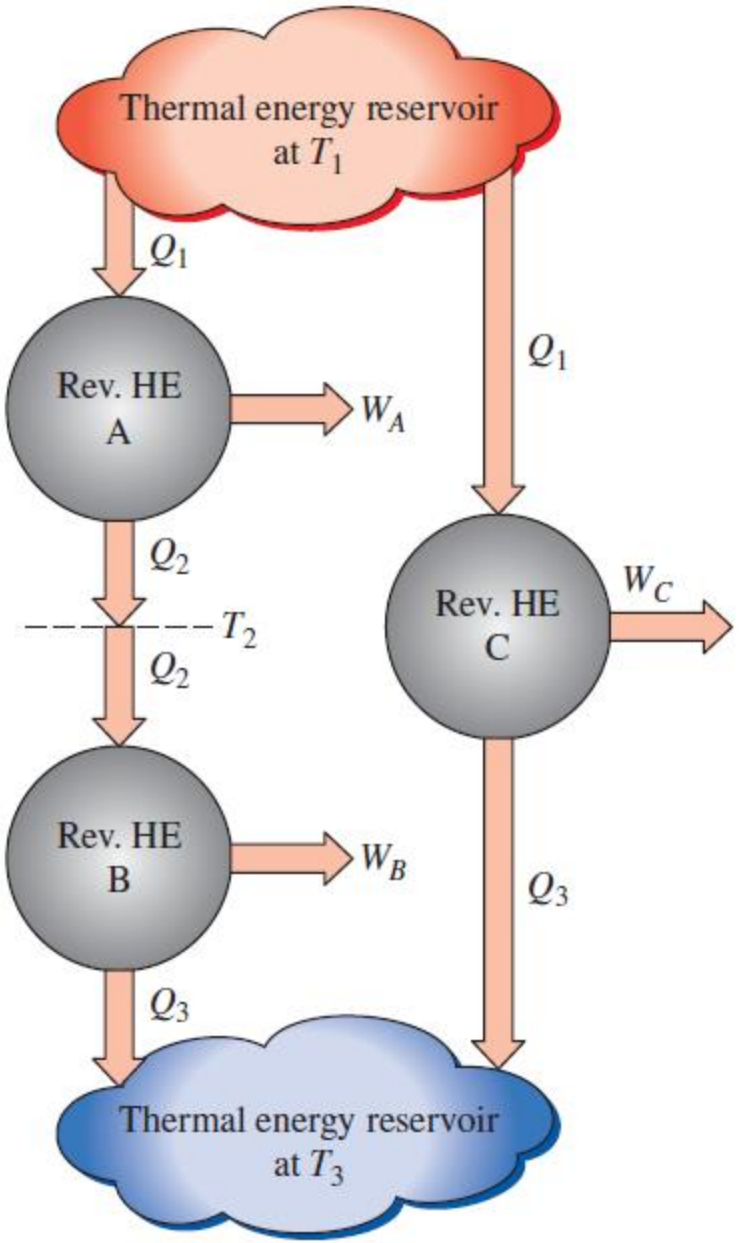


The efficiency of a reversible engine is **independent** of the **working fluid** employed and its properties, the way **the cycle** is executed, or the **type of reversible engine** used.

It depends on the same two thermal reservoirs and reservoirs are characterized by their temperatures

$$\eta_{th,rev} = g(T_H, T_L)$$

$$\frac{Q_H}{Q_L} = f(T_H, T_L)$$



$$\frac{Q_1}{Q_2} = f(T_1, T_2), \quad \frac{Q_2}{Q_3} = f(T_2, T_3), \quad \text{and} \quad \frac{Q_1}{Q_3} = f(T_1, T_3)$$

$$\frac{Q_1}{Q_3} = \frac{Q_1}{Q_2}\frac{Q_2}{Q_3}$$

$$f(T_1, T_3) = f(T_1, T_2) \cdot f(T_2, T_3)$$

$$f(T_1, T_2) = \frac{\phi(T_1)}{\phi(T_2)} \quad \text{and} \quad f(T_2, T_3) = \frac{\phi(T_2)}{\phi(T_3)}$$

$$\frac{Q_1}{Q_2} = f(T_1, T_3) = \frac{\phi(T_1)}{\phi(T_3)}$$

Absolute temperature scale

- Lord Kelvin was the first to suggest: $\left(\frac{Q_H}{Q_L}\right)_{\text{rev}} = \frac{T_H}{T_L}$

- The **temperature** in the above relationship is given on the **absolute scale**.

- The absolute temperature scale is also called as **Kelvin scale** in his honour.

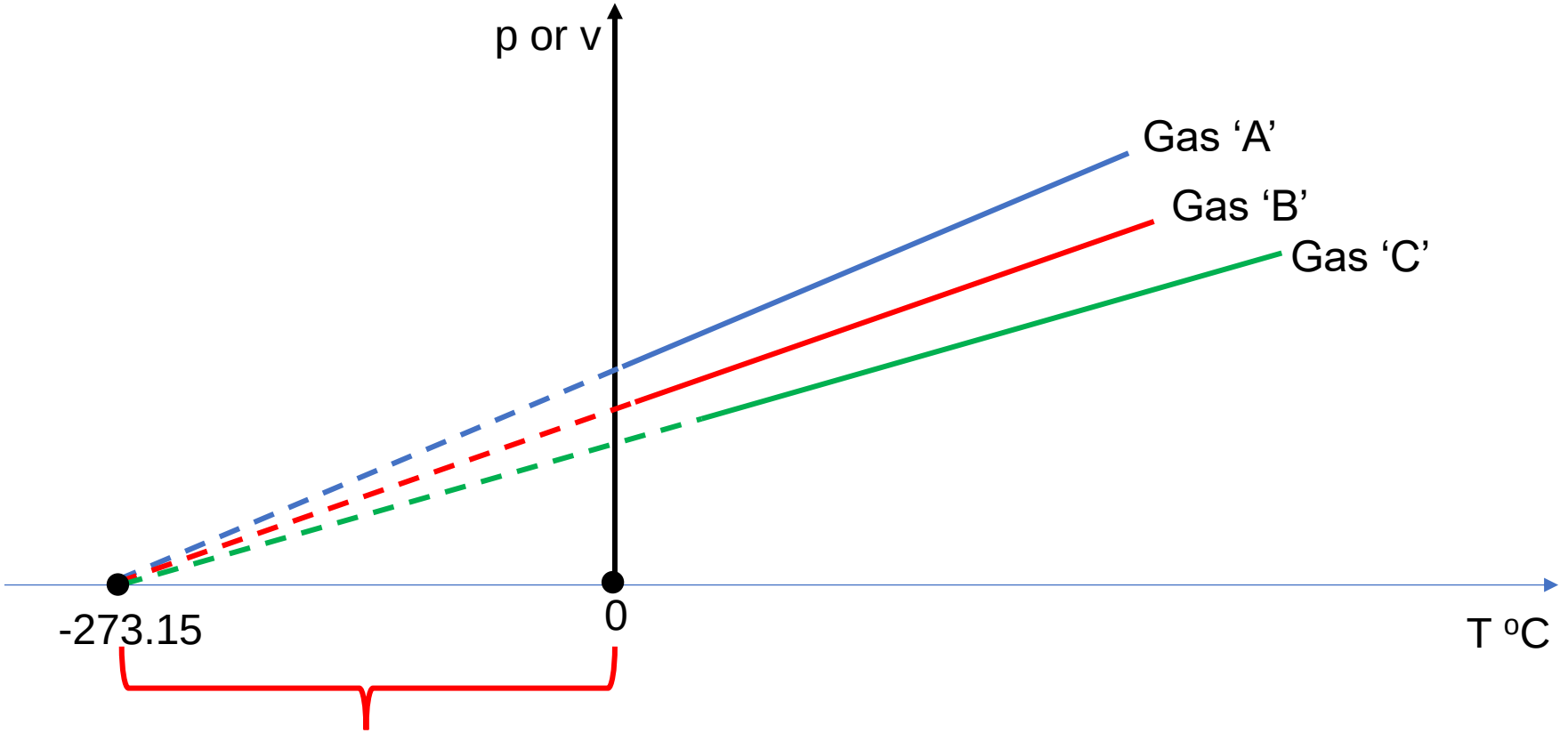
- At the International Conference on Weights and Measures held in 1954, the triple point of water was assigned the value 273.16 K. The magnitude of a Kelvin is defined as 1/273.16 of the temperature interval between absolute zero and the triple-point temperature of the water.

- The magnitudes of temperature units on the Kelvin and Celsius scales are identical (1 K ≡ 1°C). The temperatures on these two scales differ by a constant 273.15:

$$T(\,^\circ C) = T(K) - 273.15$$

Temperature scale obtained by gas thermometers

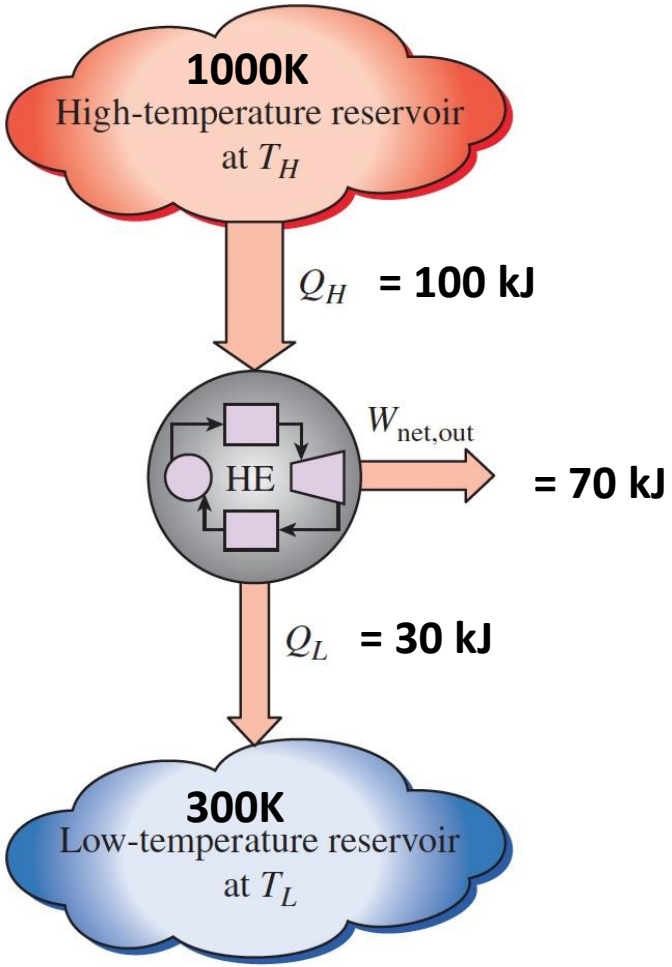
- Temperature scales obtained by constant pressure or constant volume gas thermometers using ideal gases



- Irrespective of the ideal gas, X-intercept of -273.15 was obtained

Reversible Heat Engine

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_H} \quad \text{or} \quad \eta_{\text{th}} = 1 - \frac{Q_L}{Q_H}$$



Based on heat $\eta_{th} = 1 - \frac{30}{100} = 0.7$

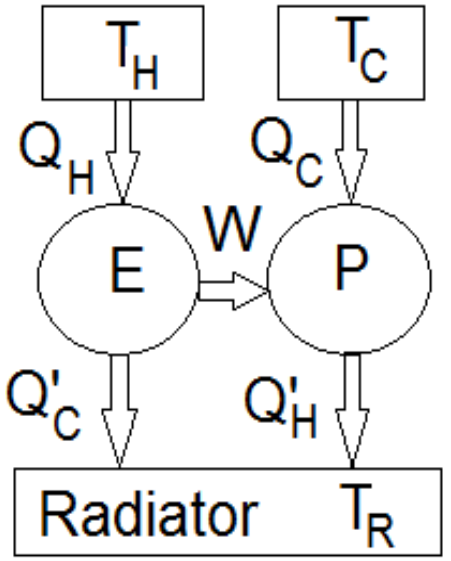
Based on temperatures of reservoirs

$$\eta_{th} = 1 - \frac{300}{1000} = 0.7$$

A heat engine is used to drive a heat pump. The heat transfers from the heat engine and from the heat pump are used to heat water circulating through the radiators of a building. The efficiency of the heat engine is 29% and the COP of the heat pump is 3.5. Evaluate the ratio of heat transfer to the radiator water to the heat transfer to the heat engine.

Given that, $\eta=0.29$ and COP = 3.5

To find: $(Q'_C + Q'_H)/Q_H$



By I law, $Q_H - Q'_C = W = Q'_H - Q_C$

Efficiency of the engine, $\eta = 1 - Q'_C/Q_H = 0.29$

==> $Q'_C/Q_H = 0.71$

COP of the pump = $3.5 = Q'_H/(Q'_H - Q_C)=1/(1- Q_C/Q'_H)$

==> $Q_C/Q'_H =0.7143$

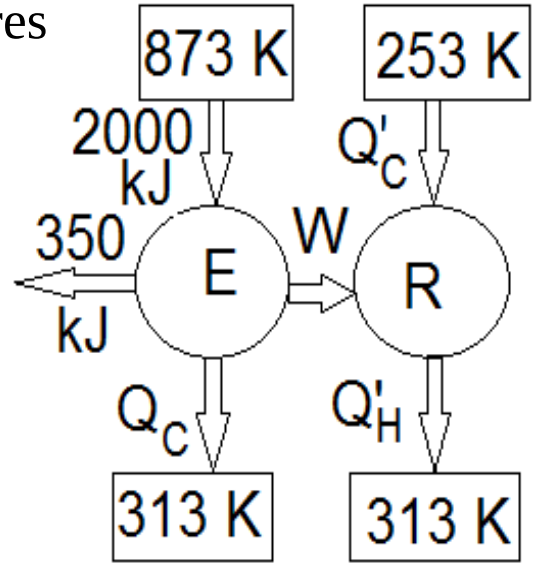
By I law, $0.29Q_H = Q'_H - Q_C$ or, $0.29Q_H/Q'_H = 1- Q_C/Q'_H$

==> $Q'_H/Q_H = 1.015$

$(Q'_C + Q'_H)/Q_H$

$= 1.015+0.71 = 1.725$

A reversible heat engine operates between two systems at constant temperatures of 600°C and 40°C. The engine drives a reversible refrigerator, which operates between systems at constant temperatures of 40°C and -20°C. The heat transfer to the engine is 2000 kJ and the net work output of the combined engine-refrigerator is 350 kJ. Evaluate the heat transfers involving the refrigerator.



Since the engine is reversible, its efficiency can be evaluated using the absolute temperatures of hot and cold reservoirs as
$\eta = 1 - 313/873 = 0.6415$.

From the first law, the total work delivered by the engine,
$W_E = \eta Q_H = 0.6415 \times 2000 = 1283$ kJ.

Therefore, the work supplied to the reversible refrigerator is $W = 1283 - 350 = 933$ kJ.
$Q'_H - Q'_C = 933$ kJ.

COP of the reversible refrigerator is given in terms of hot and cold absolute temperatures as:

$Q'_C/W = Q'_C/(Q'_H - Q'_C) = 1/[(313/253) - 1] = 4.2167$
==> $\mathbf{Q'_C} = 4.2167 \times 933 =$ **3934.15 kJ**

$\mathbf{Q'_H} = 3934.15 + 933 =$ **4867.15 kJ**.

A Carnot engine operates between two reservoirs at temperatures of T_H K and T_C K. The work output of the engine is 0.6 times the heat rejected. Given that the difference in temperature between the source and sink is 200°C, calculate (i) the source temperature, (ii) the sink temperature, and (iii) thermal efficiency.

Given $W = 0.6Q_L$, 0.6 times the heat rejected and $T_H - T_C = 200$ K.

Since, the Carnot engine is reversible,

its efficiency is determined by $\eta = 1 - (T_L/T_H) = 200/T_H$.

Work, $W = Q_H - Q_L = 0.6Q_L$.

This implies,

$Q_H/Q_L - 1 = 0.6$ or $Q_H/Q_L = T_H/T_L = 1.6 \Rightarrow 1.6T_L - T_L = 200$

or T_L =**333.33 K** and T_H =**533.33 K**

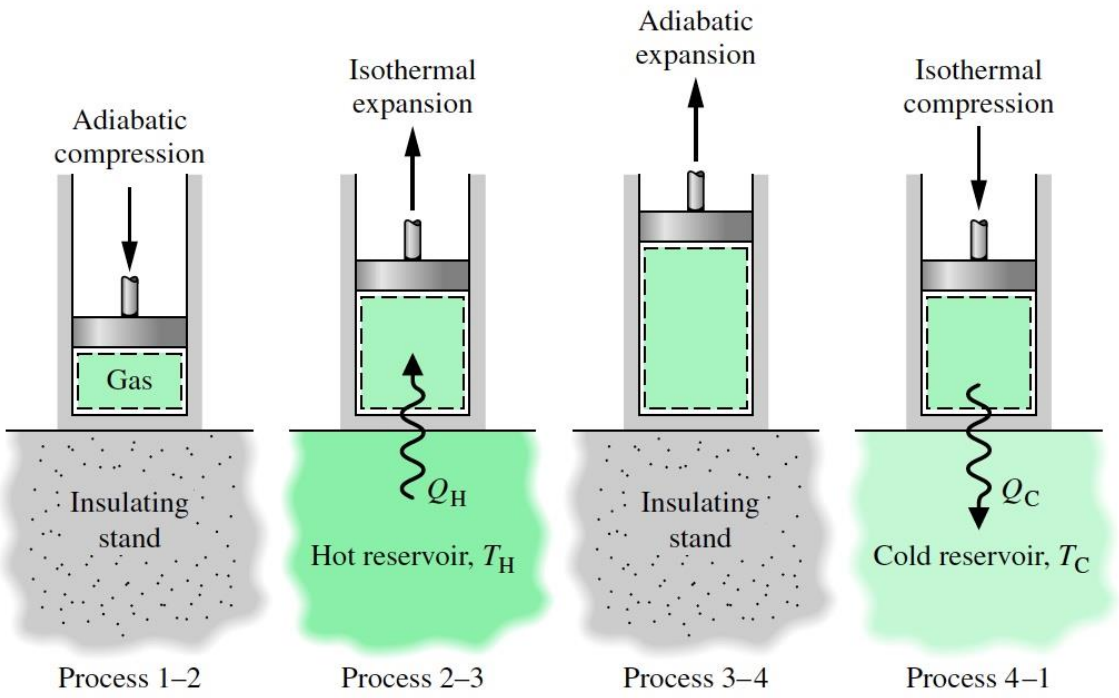
The thermal efficiency = 200/533.33 = 0.375

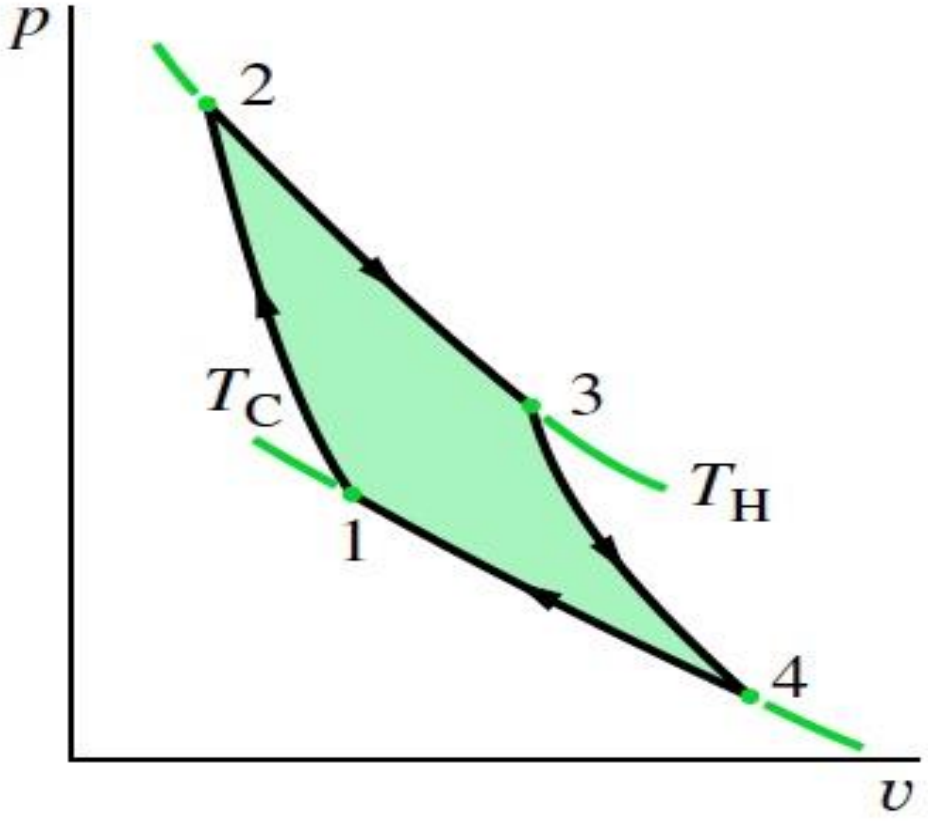
Carnot Cycle

- Carnot cycle is a thermodynamic cycle that avoids all irreversible processes
- Simple Carnot cycle comprising isothermal processes followed by adiabatic processes
- Heat transfer processes (heat addition and heat rejection) also reversible, using isothermal heat transfer
- Since all processes are reversible, the heat engine would be reversible heat engine yielding the maximum efficiency for an engine operating between the given pair of source/sink reservoirs

Carnot Cycle

- 1-2: Adiabatic compression
- 2-3: Isothermal heat addition from a reservoir at temperature T_H
- 3-4: Adiabatic expansion
- 4-1: Isothermal heat rejection to a reservoir at temperature T_C





Carnot cycle analysis assuming that the working fluid is an ideal gas

Process	Equation	Q	W	ΔU
1 – 2 (adiabatic)	$P_1V_1^{\gamma} = P_2V_2^{\gamma}$	0	$\frac{P_1V_1 - P_2V_2}{\gamma - 1}$	$-\left(\frac{P_1V_1 - P_2V_2}{\gamma - 1}\right)$
2 – 3 (isothermal)	$P_2V_2 = P_3V_3$	$P_2V_2 \ln\frac{V_3}{V_2}$	$P_2V_2 \ln\frac{V_3}{V_2}$	0
3 – 4 (adiabatic)	$P_3V_3^{\gamma} = P_4V_4^{\gamma}$	0	$\frac{P_3V_3 - P_4V_4}{\gamma - 1}$	$-\left(\frac{P_3V_3 - P_4V_4}{\gamma - 1}\right)$
4 – 1 (isothermal)	$P_4V_4 = P_1V_1$	$P_4V_4 \ln\frac{V_1}{V_4}$	$P_4V_4 \ln\frac{V_1}{V_4}$	0

Net Work done $= \frac{P_1V_1 - P_2V_2}{\gamma - 1} + P_2V_2 \ln\frac{V_3}{V_2} + \frac{P_3V_3 - P_4V_4}{\gamma - 1} + P_4V_4 \ln\frac{V_1}{V_4}$

$= \frac{mRT_1 - mRT_2}{\gamma - 1} + mRT_2 \ln\frac{V_3}{V_2} + \frac{mRT_3 - mRT_4}{\gamma - 1} + mRT_4 \ln\frac{V_1}{V_4}$

Carnot cycle efficiency

$$\text{Net work} = mRT_2 \ln\frac{V_3}{V_2} + mRT_4 \ln\frac{V_1}{V_4} = mRT_2 \ln\frac{P_2}{P_3} - mRT_1 \ln\frac{P_1}{P_4}$$

$$\text{Heat Supplied} = P_2V_2 \ln\frac{V_3}{V_2} = mRT_2 \ln\frac{V_3}{V_2} = mRT_2 \ln\frac{P_2}{P_3}$$

$$\text{Efficiency} = \frac{mRT_2 \ln\frac{P_2}{P_3} - mRT_1 \ln\frac{P_1}{P_4}}{mRT_2 \ln\frac{P_2}{P_3}} = 1 - \frac{T_1 \ln\frac{P_1}{P_4}}{T_2 \ln\frac{P_2}{P_3}}$$

Carnot cycle efficiency

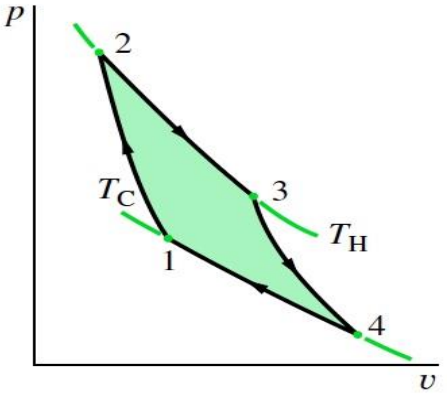
$$\frac{P_2}{P_3} = \frac{V_3}{V_2} = \frac{V_3}{V_4} \cdot \frac{V_4}{V_1} \cdot \frac{V_1}{V_2} = \left(\frac{P_4}{P_3}\right)^{\frac{1}{\gamma}} \cdot \frac{P_1}{P_4} \cdot \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}}$$

$$\left(\frac{P_2}{P_3}\right)^{1-\frac{1}{\gamma}} = \left(\frac{P_1}{P_4}\right)^{1-\frac{1}{\gamma}}$$

$$\left(\frac{P_2}{P_3}\right) = \left(\frac{P_1}{P_4}\right)$$

Processes
$P_1V_1^{\gamma} = P_2V_2^{\gamma}$
$P_2V_2 = P_3V_3$
$P_3V_3^{\gamma} = P_4V_4^{\gamma}$
$P_4V_4 = P_1V_1$

$$\text{Carnot Efficiency} = 1 - \frac{T_1 \ln \frac{P_1}{P_4}}{T_2 \ln \frac{P_2}{P_3}} = 1 - \frac{T_1}{T_2} = 1 - \frac{T_C}{T_H}$$



Thank You